

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY.

DIGITAL IMAGE PROCESSING

MODEL QUESTION (EL)

(Maximum Marks :100)

PART – A

(Answer all questions in one or two sentences. Each question carries 2 marks).

I

1. Give any two applications of Digital image processing .
2. State the four neighbor co- ordinates of a pixel.
3. Define image enhancement.
4. Give any three redundancies which are used in image compression.
5. Draw a model for an image degradation process.

(5x2)

(Answer any five of the following questions . Each question carries 6 marks).

PART - B

II

1. Describe the various steps required in Digital Image Processing.
2. Explain neighbors and connectivity of pixels in an image.
3. Describe the functions of Fast Fourier Transform in image processing.
4. State the functions of High pass filter in image enhancement.
5. Describe Image Compression standard – JPEG
6. Explain the method for line detection in image segmentation.
7. State the concept of restoration in image processing

(5x6)

PART – C

(Answer one full question from each module . Each question carries 15 marks).

III

- (a) Describe the basic blocks of Digital Image Processing. (9)
- (b) Describe Sampling and quantization in Digital Image Processing (6)

OR

IV

- (a) Explain the methods of distance measure between any two pixels (8)
- (b) Explain the important aspects of Digital image representation (7)

V

- (a) Describe Discrete cosine and Haar image transforms (8)
- (b) Explain negative of an image in image enhancement (7)

OR

VI

- (a) Give the tree diagram of image enhancement techniques (8)
- (b) Describe contrast stretching (7)

VII

- (a) Describe Huffman coding for image compression (8)
- (b) Explain lossless predictive coding (7)

OR

VIII

- (a) Describe Threshold coding for image compression (8)
- (b) Describe JPEG image compression standard (7)

IX

- (a) Explain the method of edge detection for image segmentation (9)
- (b) Describe basic concepts of pattern recognition using image processing (6)

OR

X

- (a) Describe region splitting and merging for image segmentation (8)
- (b) Describe the method for detection of isolated points in segmentation (7)